

IMAGE RECLASSIFICATION

Creating Subset

1. Add raster > “Landsat8_2021_R_clip.tif”
2. Add vector > “boundary.shp”
3. Go to “Raster > Extraction > Clip Raster by Mask Layer”
4. Select “Input layer” as “Landsat8_2021.tif” from the dropdown list
5. Provide the output location for “Clipped (extent)”
6. Define “Clipping extent” as calculated from layer > boundary
7. Then click “Run” button
8. Now the clipped image will open
9. The band combination of the image can be changed into false color combination (5,4,3)

Producing NDVI image

1. Go to “Raster > Raster Calculator”
2. NDVI formula can be defined as; $NDVI = (NIR - RED) / (NIR + RED)$
3. Define “Output Layer” file location and provide the file name as “CNDVI”
4. Check the “Selected Layer Extent”
5. Then click “OK” button
6. Now the NDVI layer will open
7. Please notice that the range of NDVI layer is -1 to 1
8. To change the color range, right click on the layer name and select “Properties”
9. Now open “Symbology” tab
10. Select “Singleband pseudocolor” for “Render type”
11. Select a suitable “Color ramp”
12. Then click “Classify”
13. Now the color image will open in the map view

14. Please try to distinguish high and low NDVI areas

Defining feature classes

1. To add the toolbox, go to “Processing > Toolbox”
2. Now “Processing Toolbox” will open
3. Then go to “Raster analysis” and select “Reclassify by table”
4. Select the input layer from the drop down list of “Raster layer”
5. Go to “Reclassification table” and define reclassification ranges for feature classes as follows
 - NDVI -1 – 0 – Water (Class 1)
 - NDVI 0 – 0.1 – Sand (Class 2)
 - NDVI 0.1 – 0.3 – Urban (Class 3)
 - NDVI 0.3 – 0.4 – Grass (Class 4)
 - NDVI 0.4 – 1 – Healthy vegetation (Class 5)
6. Use “Add row” button to add more rows and manually feed the ranges
10. After completing, click “OK” button
11. Change the “Output data type” into “Byte”
12. Provide the location to save the file in “Reclassified raster”
13. Then click “Run” button
14. Then the process will start
15. After the completion of the process, click “Close”
16. Then the reclassified image will open
17. To change symbology into unique values, right click on the layer name and select “Properties”
18. Now the “Layer Properties” window will open
19. Select Render type“ as “Paletted/unique values”
20. Select “Color ramp” as “Random colors”
21. Then click “Apply” and “OK”

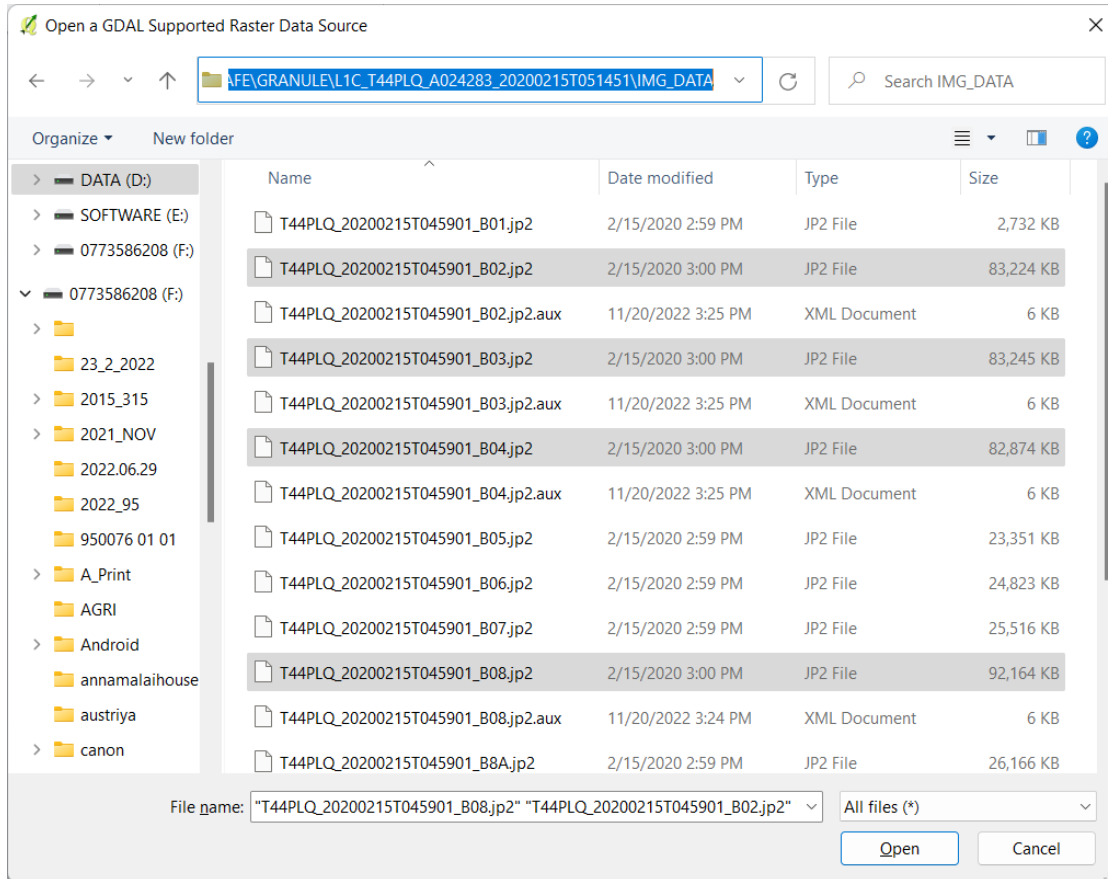
22. Now the new reclassified image will open
23. To change the color and labels in a meaningful way, go to “Layer Properties” window again
24. Edit the label name by double clicking on “Label” column
25. Also by double clicking on the color, a suitable color can be selected
26. Then click “Apply” and “OK”
27. Now the new classified image will open
28. Zoom to a location and compare the original satellite image and NDVI classified image

Converting a raster layer into a vector layer

2. Go to “Raster > Conversion > Polygonize (Raster to Vector)”
3. Select the classified image as “Input layer”
4. Then click “Run” button
5. It will take some time to process

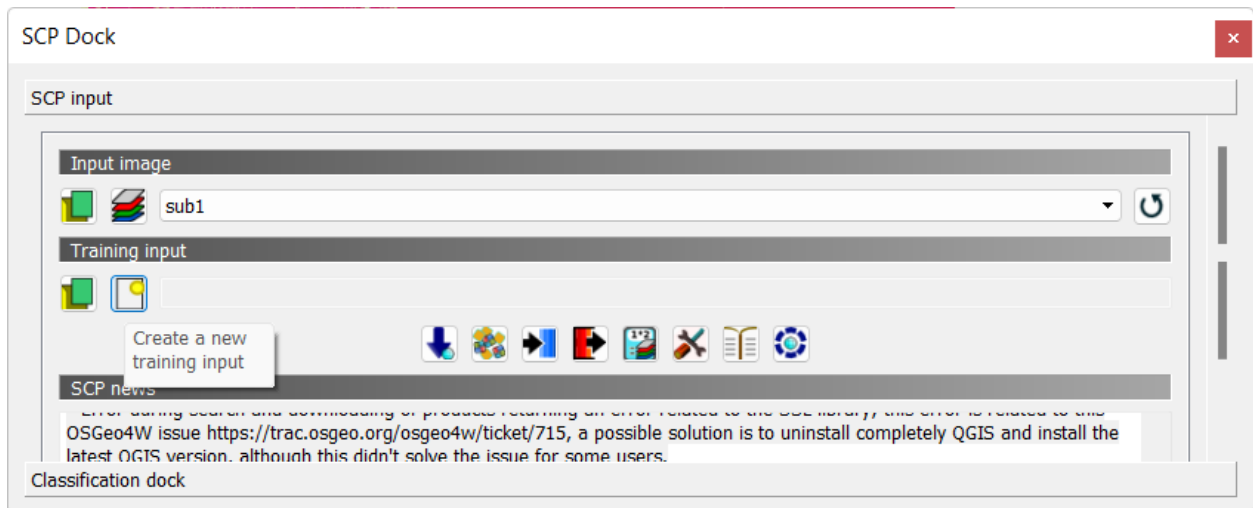
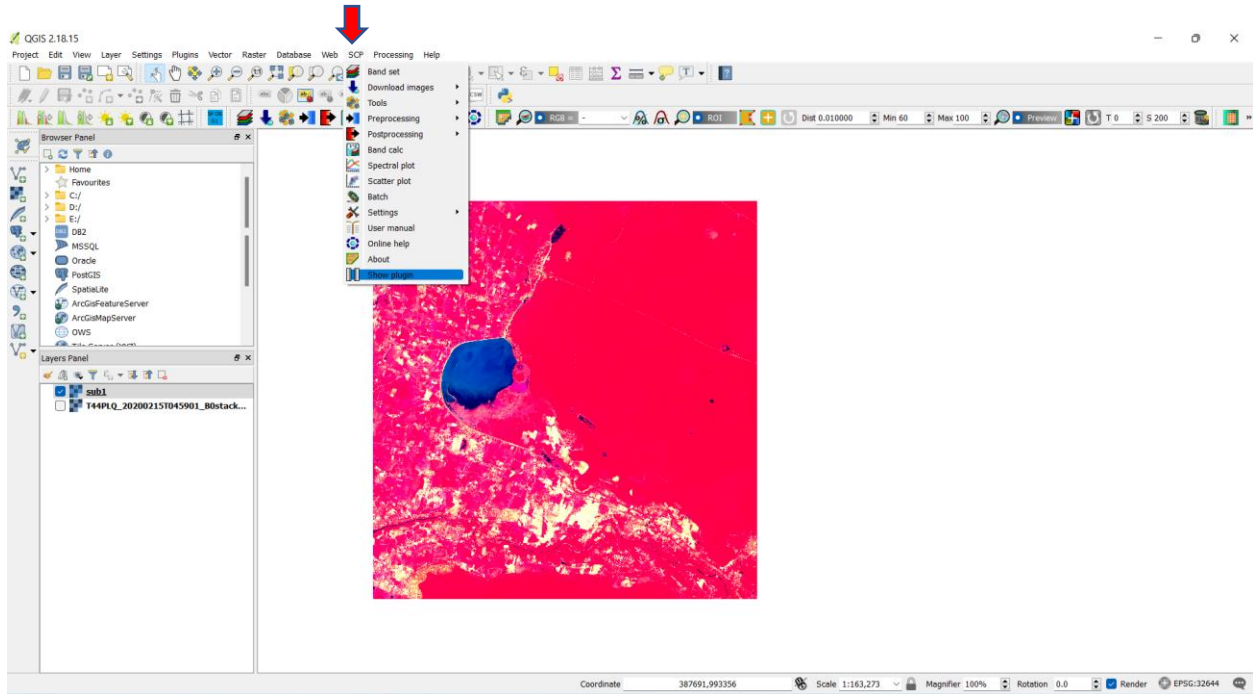
IMAGE CLASSIFICATION

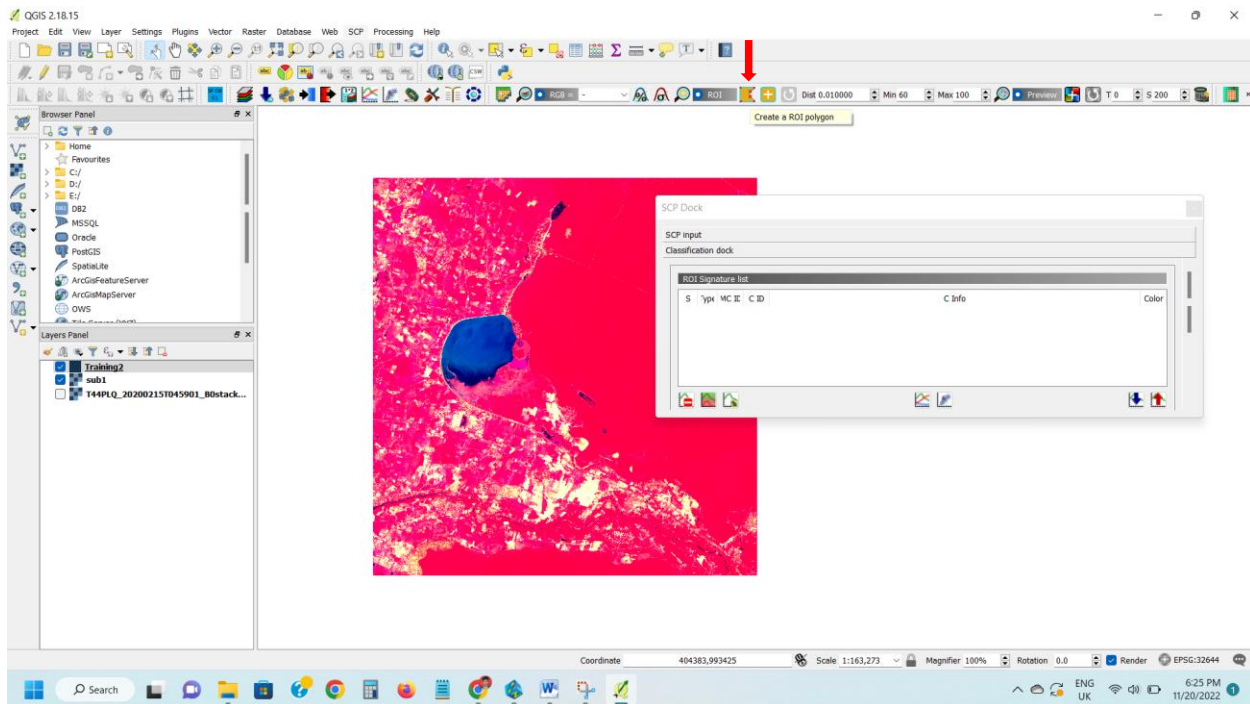
1. Go to Plugins
2. Search the “Semi-Automatic Classification” tool & Install Plugin
3. Add Raster Data set

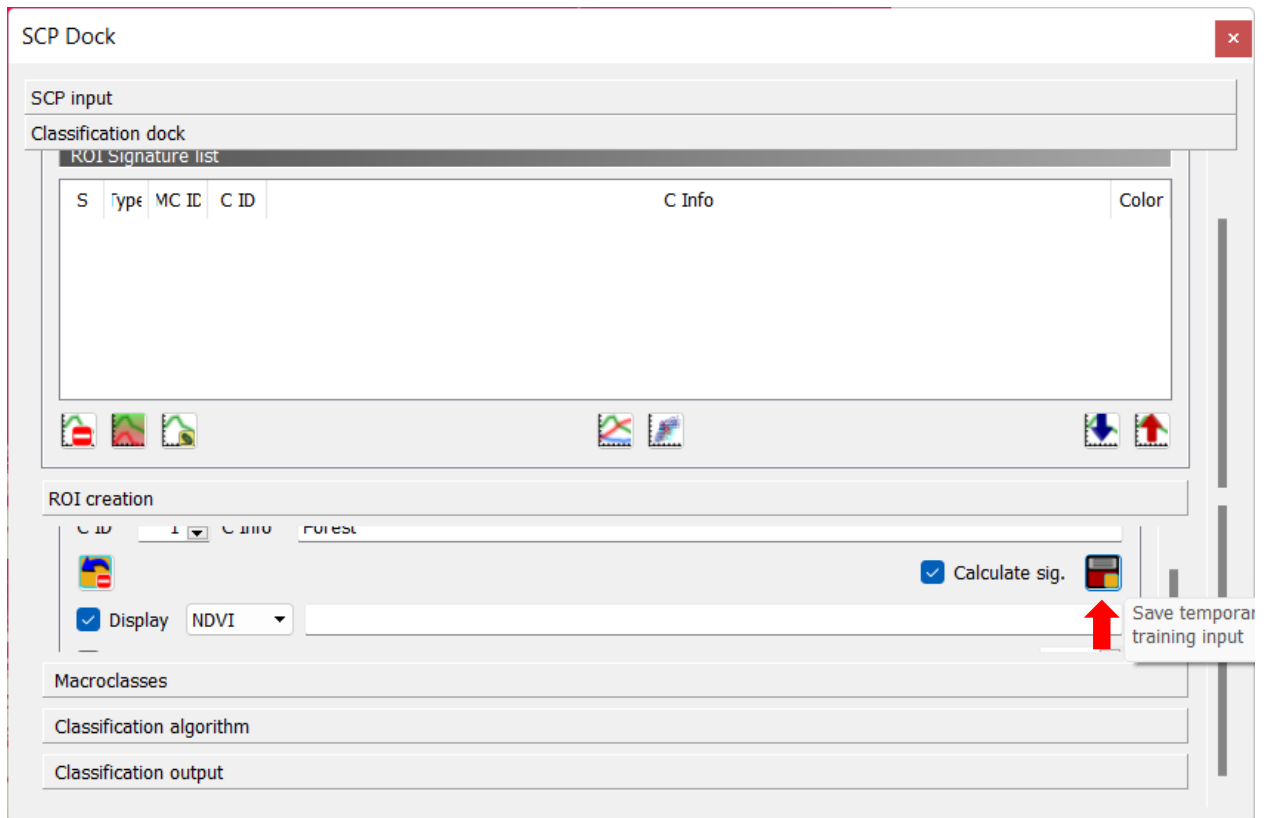
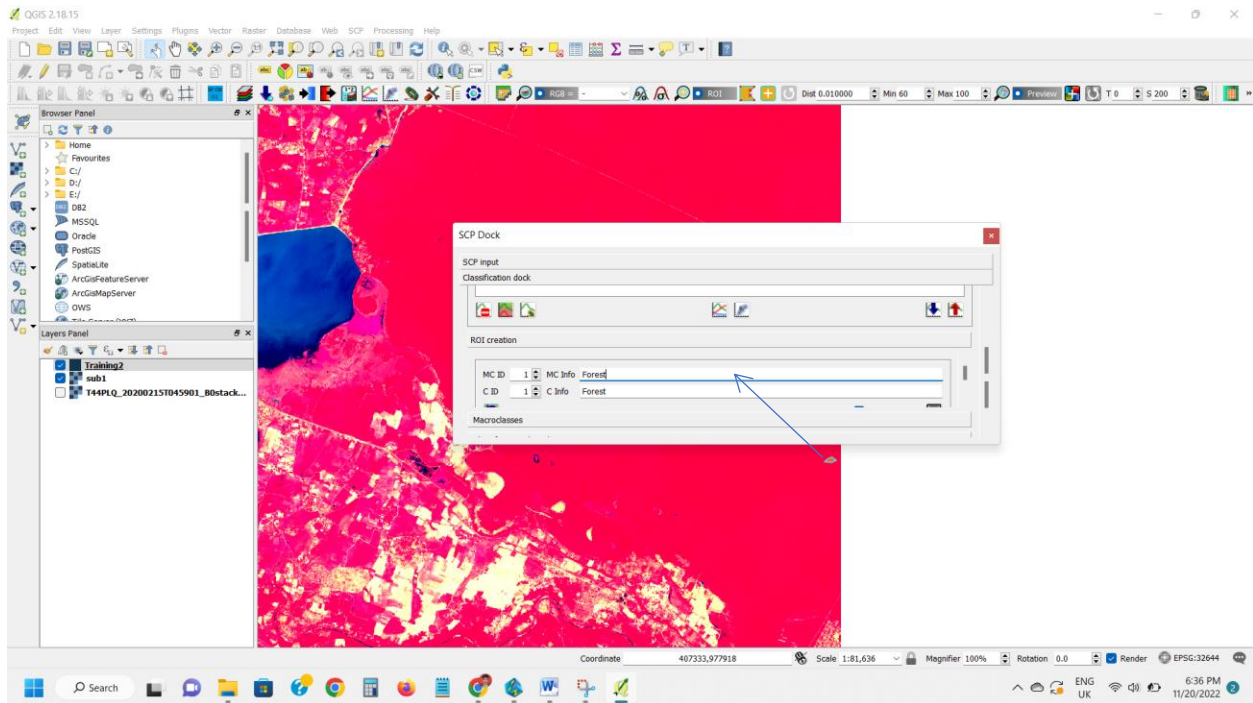


4. For create colour composite image... Go to SCP -----> Band Set
5. Make True Colour Composite image (1,2,3)
6. False colour composite image (4,3,2)
7. Clip Study Area using shapefile 'sub.shp'

Classification *****







SCP Dock



SCP input

Classification dock

ROI Signature list

	S	Type	MC ID	C ID	C Info	Color
1	☒	B	1	1	merged_Forest	
2	☒	B	2	2	merged_Water	
3	☒	B	3	3	merged_Paddy	
4	☒	B	4	4	merged_Resident	
5	☒	B	5	5	merged_Barrent	



ROI creation

Macroclasses

Classification algorithm

Use ☒ MC ID ☐ C ID

Algorithm

Maximum Likelihood



Threshold

0.0000



Land Cover Signature Classification

Use ☒ LCS☐ Algorithm☐ only overlap

Classification output

